

# Sampling Distributions

## Introduction:

In a statistical investigation, usually we study the various aspects of a population characteristic (or characteristics) of a group of individuals or things etc. (called population). If the population is large (infinite) and the complete enumeration of population is not practicable or possible, then we consider a finite subgroup or some part of the population called sample and based on the sample information, we assess characteristics of the population. This process is known as sampling. The following examples can be useful to study and understand this definition.

If we consider the example of a company producing electrical fans, it is supposed to test life time of the fans under the specifications of the company. In this example, testing (i.e. complete enumeration) of all fans is much expensive and time taking, hence they consider some electrical fans at regular intervals called samples which are used to assess or estimate the life time of electrical fans (i.e. population).

Sometimes, even if complete enumeration of the population is possible, it is not at all desirable. For example, crackers, nuclear weapons, explosive materials etc. in which the items are destroyed in examining them 100% inspection. In this example even if 100% inspection is possible and population is finite also, it is not all desirable.

In some cases, even if the population is finite and even the inspection is not destructive, 100% inspection (complete enumeration) may not be considered because of several problems viz., administrative and financial implications, time considerations etc.

For example, a car manufacturing company produces 200 cars per a day. Testing of all 200 100% inspection) produced in a day may not be considered due to and cost factors even the population is finite.

In all these cases, the concept of sampling is used to assess the population characteristics. Simply to say, sampling is quite often used in our day-to-day life. For example, to assess the quality of rice, we take handful of rice from the bag and then we decide whether to purchase or not. Sampling is the most effective technique to draw valid conclusions about population characteristics in less time with more accuracy.

## KEY DEFINITIONS

### 1. Population

The group of individuals (or) things (or) items under a statistical investigation is called population (or) universe. Population can also be defined as collection of animate and inanimate things under the study. **In other words**

A population is the complete collection of all individuals, objects, observations, or measurements that satisfy a particular condition and are of interest in a study.

In statistics, the population is the entire group about which we want to draw conclusions.

### Example 1 (Students)

Suppose a college has **5,000 students**, and you want to know the average height of the students.

- **Population:** All 5,000 students in the college.

### Example 2 (Computers)

A company manufactures **10,000 laptops** in a month.

- **Population:** All 10,000 laptops produced that month.

### Example 3 (Election)

To estimate the voting preference in a state,

- **Population:** Every eligible voter in the state.

## 2. Sample

A finite subset of a population is called a sample, i.e., some selected individuals or things from the population (total group of individuals or things) constitute a sample. (OR)

A sample is a small subset of the population selected for observation or analysis. It should represent the characteristics of the entire population.

### Example 1 (Students)

From the 5,000 students in the college, suppose you randomly select **200 students**.

- **Population:** 5,000 students
- **Sample:** 200 selected students

### Example 2 (Laptops)

Out of 10,000 laptops manufactured,

- Inspect only **300 laptops** for quality.
- These 300 laptops form the **sample**.

### Example 3 (Hospital)

A hospital wants to estimate the average blood pressure of patients.

- Total patients: 2,000
- Selected patients: 150

Here,

- Population = 2,000 patients

- Sample = 150 patients

**Samples are classified in two ways i.e., large sample and small sample.**

1. **Large sample:** If the size of the sample ( $n$ )  $\geq 30$ , the sample is said to be large sample.
2. **Small sample:** If the size of the sample ( $n$ )  $< 30$ , the sample is said to be small sample (OR) exact sample.

### **3. Sampling**

**Sampling** is the **process or technique of selecting a sample from the population** so that conclusions about the population can be made without studying every individual.

Sampling saves **time, money, and effort**.

#### **Example 1 (Opinion Poll)**

A city has **10 lakh voters**.

Instead of asking all 10 lakh voters,

- Survey **2,000 randomly selected voters**.

The process of selecting these 2,000 voters is called **sampling**.

#### **Example 2 (Quality Control)**

A biscuit factory produces **50,000 biscuits every day**.

Checking every biscuit is impossible.

Instead,

- Inspect **500 biscuits** chosen randomly.

This selection process is called **sampling**.

### **4. Parameter**

A parameter is a numerical measure that describes a characteristic of an entire population. Since it involves every member of the population, it is usually fixed but often unknown.

Common Population Parameters

1. Population mean:  $\mu$
2. Population standard deviation:  $\sigma$
3. Population proportion:  $P$

Examples:

1. Average annual sales of a product in a company.

2. The standard deviation of sales of a product.
3. The variance of paddy consumption in the state Telangana.

## 5. Statistic

A **statistic** is a **numerical measure calculated from a sample**. It is used to estimate the corresponding population parameter.

### Common Sample Statistics

1. Sample mean:  $\bar{x}$
2. Sample standard deviation:  $s$
3. Sample proportion:  $p$

### Example 1: Students' Heights

From the 5,000 students, a researcher randomly selects **200 students**.

The average height of these 200 students is **167.5 cm**.

Sample: 200 students

Statistic (Sample Mean):  $\bar{x} = 167.5cm$

This value is a **statistic** because it is calculated from a sample.

### Example 2: Laptop Quality

A company manufactures **10,000 laptops**.

The quality department inspects **300 laptops**.

The average battery life of these 300 laptops is **9.8 hours**.

Population(N): 10,000 laptops

Sample(n): 300 laptops

Statistic:  $\bar{x} = 9.8 \text{ hours}$

## 6. Sampling Methods

Sampling methods are techniques used to select a subset (sample) from a population so that conclusions can be made about the entire population.

### 1. Simple Random Sampling (SRS)

#### Definition:

**Simple Random Sampling** is a sampling technique in which **every member of the population has an equal and independent chance of being selected**.

In this method, selection is completely based on chance, like drawing names from a lottery.

There are two types of Simple Random Samplings (SRS) i.e., Simple Random Sampling Without Replacement (SRSWOR) and Simple Random Sampling with Replacement (SRSWR)

### 1. Simple Random Sampling Without Replacement (SRSWOR)

**Definition:** Simple Random Sampling Without Replacement (SRSWOR) is a sampling method in which once a unit is selected, it is not returned to the population before selecting the next unit. Therefore, the same unit cannot be selected more than once.

If the population size is 'N' and the sample size is 'n' then the number of possible samples drawn from the population without replacement is  $N^n$

#### Characteristics

- Every unit has an equal chance of being selected.
- A selected unit is **not replaced**.
- No duplicate selections are possible.
- The probability of selection changes slightly after each draw.

### 2. Simple Random Sampling with Replacement (SRSWR)

**Definition:** Simple Random Sampling with Replacement (SRSWR) is a sampling method in which each selected unit is returned to the population before the next selection. Hence, the same unit can be selected more than once.

If the population size is 'N' and the sample size is 'n' then the number of possible samples drawn from the population with replacement is  $N_{c_n}$

#### Characteristics

- Every unit has an equal chance of selection in every draw.
- The selected unit is **replaced** before the next draw.
- Duplicate selections are possible.
- The probability remains the same for each draw.

## 2. Stratified Sampling

**Definition:** Stratified Sampling is a sampling technique in which the population is divided into homogeneous groups called strata, and then a simple random sample is selected from each stratum.

This ensures that **every important subgroup is represented** in the sample.

**Example 1: College Students:** A college has **1,000 students**.

### Department Students

CSE            500

ECE            300

Mechanical 200

Suppose a sample of **100 students** is needed.

Select students proportionally from each department:

- CSE → 50 students
- ECE → 30 students
- Mechanical → 20 students

Then use **simple random sampling** within each department.

### Example 2: Employee Survey

A company has:

- Managers = 100
- Engineers = 300
- Technicians = 600

To survey 100 employees, select respondents proportionally from each group so that all categories are represented.

## 3. Systematic Sampling

**Definition:** Systematic Sampling is a sampling technique in which every  $k^{\text{th}}$  member of the population is selected after choosing a random starting point.

The sampling interval is calculated as:  $k = \frac{N}{n}$

where:

- N = Population size
- n = Sample size
- k = Sampling interval

### Example 1: Students

A college has **1,000 students**.

A sample of **100 students** is required.

Sampling interval:

$$k=1000/100=10$$

Choose a random starting number between 1 and 10, say **7**.

The selected students are:

7, 17, 27, 37, 47, ...

Every 10th student is included in the sample.

### Example 2: Manufacturing

A factory produces **2,000 bulbs**.

The quality inspector wants a sample of **200 bulbs**.

Sampling interval:

$$k=2000/200=10$$

Choose a random starting point, then inspect every 10th bulb.

### Comparison of Sampling Methods

| Feature             | Simple Random Sampling  | Stratified Sampling                            | Systematic Sampling                     |
|---------------------|-------------------------|------------------------------------------------|-----------------------------------------|
| Selection method    | Completely random       | Random selection within each stratum           | Every kth unit after a random start     |
| Population division | No                      | Yes (into strata)                              | No                                      |
| Equal chance        | Yes                     | Yes, within each stratum                       | Yes, after the random start             |
| Best suited for     | Homogeneous populations | Heterogeneous populations with distinct groups | Ordered lists without periodic patterns |
| Example             | Lottery method          | Department-wise student selection              | Every 10th student                      |

### Real-Life Example (Hospital)

A hospital has **2,000 patients**.

- **Simple Random Sampling:** Randomly select 200 patients from the full list.
- **Stratified Sampling:** Divide patients into children, adults, and senior citizens, then randomly sample from each age group.

- **Systematic Sampling:** Choose a random starting patient, then select every 10th patient from the registration list.

**Another example:** A cloud company has **10,000 servers**.

- **Simple Random Sampling:** Randomly choose 500 servers for performance testing.
- **Stratified Sampling:** Divide servers into database, application, and web servers, then randomly sample from each category.
- **Systematic Sampling:** After a random start, monitor every 20th server in the inventory.

**Easy Way to Remember**

- **Simple Random Sampling:** *Everyone has an equal chance.*
- **Stratified Sampling:** *Divide into groups first, then sample from each group.*
- **Systematic Sampling:** *Choose a random start, then select every  $k^{th}$  member.*

## 7. Sampling Distribution

The number of all possible samples of size  $n$  drawn from a population of size  $N$  is a statistic say  $t$  (e.g. may be mean, variance, S.D. etc.) can be computed for each and every sample, which will vary from sample to sample. The aggregate of the various values of the statistic under consideration so obtained, may be grouped into a frequency distribution is known as sampling distribution of the statistic.

Examples:

let  $X_1, X_2, \dots, X_n$  be a sample drawn from a normal population whose mean is  $\mu$  and variance  $\sigma^2$ . If we compute the statistic as the sample mean ( $\bar{X}$ ), then the sample mean follows normal distribution with mean  $\mu$  and variance  $\frac{\sigma^2}{n}$ .

## 8. Sampling Distribution of a Statistic

**Definition:**

A sampling distribution of a statistic is the probability distribution of all possible values of a statistic (such as the sample mean, sample proportion, or sample variance) obtained from all possible random samples of the same size drawn from a population.

In simple words, it describes **how a statistic varies from one random sample to another**.

**Simple Definition:** A sampling distribution is the distribution formed by calculating a statistic for every possible sample of a fixed size taken from a population.

The main characteristic of the Sampling distribution of a statistic is that it approaches normal distribution even when the population distribution is not normal provided the sample size is sufficiently large (greater than 30). Another important feature of the sampling distribution of

statistic is that the mean and the standard deviation of the sampling distribution of sample mean bear a definite relation to the corresponding parameters i.e., mean and standard deviation of parent population. These characteristics of the sampling distribution help us:

- (i) To estimate the unknown population parameter from the known statistic
- (ii) To set the confidence limits of the parameter within which the parameter values expected to lie.
- (iii) To test a hypothesis and to draw a statistical inference from it.

## 9. Standard Error (S. E.) of a Statistic:

The standard error of a statistic t (i.e., S. E. of sample mean, or sample S.D.) is the standard deviation of the sampling distribution of the statistic. Thus S.E. of sample mean is the S.D. of the sampling distribution of sample mean. It is used for assessing the difference between the expected value and observed value. It plays an important role in large sample theory and forms the basis in tests of hypothesis or tests of significance. It gives an idea about the reliability and precision of a sample. S. E. enables us to determine the confidence limits within which the parameters are expected to lie.

The standard errors (S.E.) of some of the well-known statistics are given below (without proof), where  $\mu$  the population mean,  $\sigma^2$  the population variance, P the population proportion, n is the sample size,  $\bar{x}$  the sample mean,  $s^2$  the sample variance and p the sample proportion.

| S. No | Statistic                                                    | Standard Error (S.E) of statistic                          |
|-------|--------------------------------------------------------------|------------------------------------------------------------|
| 1     | Sample mean ( $\bar{x}$ )                                    | $\frac{\sigma}{\sqrt{n}}$                                  |
| 2     | Sample standard deviation(s)                                 | $\frac{\sigma}{\sqrt{2n}}$                                 |
| 3     | Sample proportion(p)                                         | $\sqrt{\frac{P Q}{n}}, \text{ where } Q = 1 - P$           |
| 4     | Difference of two sample means ( $\bar{x}_1 - \bar{x}_2$ )   | $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$   |
| 5     | Difference of two sample proportions ( $p_1 - p_2$ )         | $\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$         |
| 6     | Difference of two sample standard deviations ( $s_1 - s_2$ ) | $\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$ |

For a finite population of size N, when a sample is drawn without replacement, we have

$$(i) \quad \text{S. E. of sample mean} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

$$(ii) \quad \text{S. E. of sample proportion} = \frac{PQ}{n} \cdot \sqrt{\frac{N-n}{N-1}}$$

Here, the factor  $\left(\frac{N-n}{N-1}\right)$ , often called the *finite population correction factor*.

We note that this term tends to become closer and closer to unity as population size becomes larger and larger.

### 10. Sampling Distribution of Mean ( $\sigma$ known)

The probability distribution of  $\bar{X}$  is called the sampling distribution of means. The sampling distribution of a statistic depends on the size of the population, the size of the samples, and the method of choosing the samples.

Let  $X_1, X_2, \dots, X_n$  be the random samples drawn from a population of size N with mean  $\mu$  and variance  $\sigma^2$  and  $\bar{X}$  is the mean of samples.

Infinite population: Suppose the samples are drawn from an infinite population i.e.,  $n \rightarrow \infty$  (or) sampling is done with replacement, then

- The mean of the sampling distribution of means,  $\mu_{\bar{x}} = \mu$  i.e.,  $E(\bar{X}) = \mu$
- The variance of the sampling distribution of means,  $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$  i.e.,  $Var(\bar{X}) = \frac{\sigma^2}{n}$
- The S. D. of the sampling distribution of mean,  $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

### 11. Central Limit Theorem:

If  $\bar{X}$  be the mean of a random sample of size n drawn from a population having mean  $\mu$  and S.D.  $\sigma$ , then the sampling distribution of the sample mean  $\bar{x}$  is approximately a normal distribution with mean  $\mu$  and S. D. = S.E. of  $\bar{x} = \frac{\sigma}{\sqrt{n}}$  provided the sample size is large ( $n \geq 30$ )

This is established by central limit theorem stated below (without proof)

**Theorem:** If  $\bar{x}$  be the mean of a sample size n drawn from a population with mean  $\mu$  and S. D.  $\sigma$  then the standardized sample mean

$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$  is a random variable whose distribution function approaches that of the standard

normal distribution  $N(z; 0, 1)$  as  $n \rightarrow \infty$ .